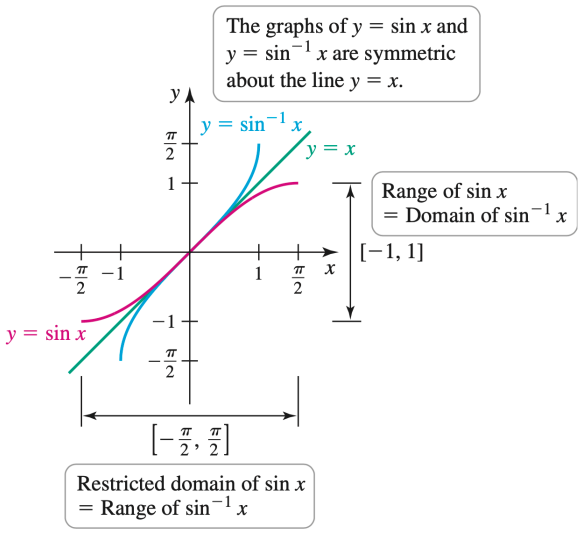
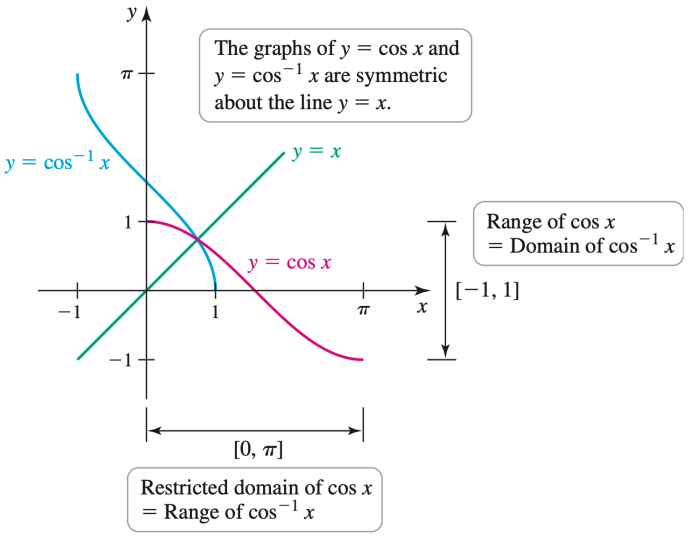
Inverse Trig Functions Review

Domain and range of sine and cosine, and their inverses:

- $\sin(x)$ domain: $[-\frac{\pi}{2}, \frac{\pi}{2}]$, range: $[-1,1]$; $\sin^{-1}(x)$ domain: $[-1,1]$, range: $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- $\cos(x)$ domain: $[0,\pi]$, range: $[-1,1]$; $\cos^{-1}(x)$ domain: $[-1,1]$, domain: $[0,\pi]$

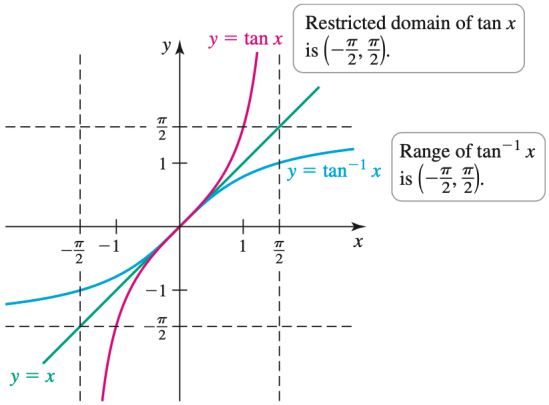
The graphs are shown below:

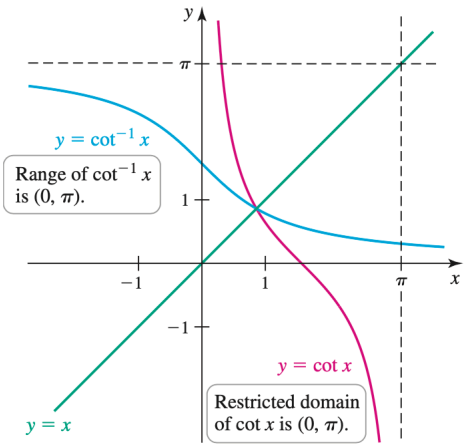




Domain and range of $\tan(x)$ and $\cot(x)$ and their inverses

- $\tan(x)$ domain: $(-\frac{\pi}{2}, \frac{\pi}{2})$, range $(-\infty,\infty)$; $\tan^{-1}(x)$ domain $(-\infty,\infty)$, range: $(-\frac{\pi}{2}, \frac{\pi}{2})$
- $\cot(x)$ domain: $(0,\pi)$, range: $(-\infty,\infty)$; $\cot^{-1}(x)$ domain: $(-\infty,\infty)$, range $(0,\pi)$

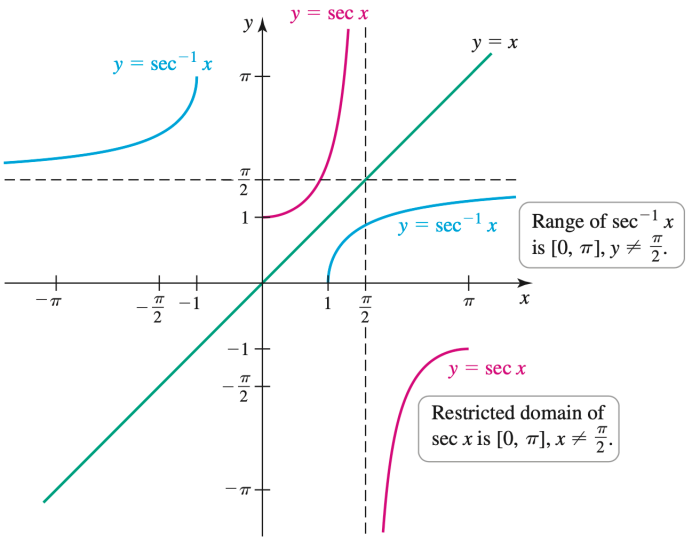


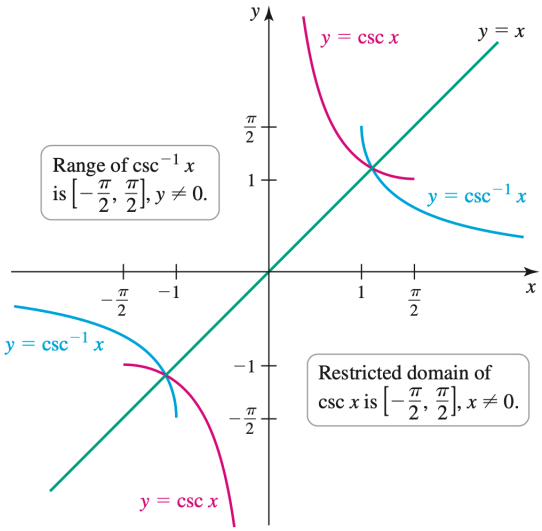


Domain and range of $\sec(x)$ and $\csc(x)$ and their inverses:

- $\sec(x)$ domain: $[0,\frac{\pi}{2})\cup(\frac{\pi}{2},\pi]$ range: $(-\infty,-1]\cup[1,\infty)$; $\sec^{-1}(x)$ domain: $(-\infty,-1]\cup[1,\infty)$, range: $[0,\frac{\pi}{2})\cup(\frac{\pi}{2},\pi]$
- $\csc(x)$ domain: $[-\frac{\pi}{2},0)\cup(0,\frac{\pi}{2}]$, range: $(-\infty,-1]\cup[1,\infty)$; $\csc^{-1}(x)$ domain: $(-\infty,-1]\cup[1,\infty)$, range: $[-\frac{\pi}{2},0)\cup(0,\frac{\pi}{2}]$

The graphs are shown below:





We need to determine where the derivatives of the 6 inverse trig functions exist, and find formulas for their derivatives. Fortunately, the following identities cut our workload in half.

Theorem.

$$\sin^{-1}(x)+\cos^{-1}(x)=\frac{\pi}{2}, \quad x\in[-1,1]$$
$$\tan^{-1}(x)+\cot^{-1}(x)=\frac{\pi}{2}, \quad x\in\mathbb{R}$$
$$\sec^{-1}(x)+\csc^{-1}(x)=\frac{\pi}{2}, \quad |x|\geq 1.$$

Proof. We prove the first identity and leave the others as an exercise. For $x\in[-1,1]$, let $y=\sin^{-1}(x)\in[-\frac{\pi}{2},\frac{\pi}{2}]$. Then $x=\sin(y)$, so we can write

$$\sin^{-1}(x)+\cos^{-1}(x)=y+\cos^{-1}(\sin(y))$$

Since $\sin(y)=\cos(\frac{\pi}{2}-y)$ (note that $y\in[-\frac{\pi}{2},\frac{\pi}{2}]$ implies $\frac{\pi}{2}-y\in[0,1]$, so it is indeed in the domain of the restricted cosine function), this becomes

$$\begin{aligned}\sin^{-1}(x)+\cos^{-1}(x)&=y+\cos^{-1}(\sin(y))\\&=y+\cos^{-1}(\cos(\frac{\pi}{2}-y))\\&=y+\frac{\pi}{2}-y\\&=\frac{\pi}{2}.\end{aligned}$$

The proofs of the remaining identities are similar, if one recalls that $\cot(\frac{\pi}{2}-y)=\tan(y)$ and $\csc(\frac{\pi}{2}-y)=\sec(y)$. □

1. Prove the second identity above.

2. Differentiate both sides of each identity above to find a relationship between the derivatives of each pair of inverse trig functions. Write these relationships below.

Recall the following theorem from last time:

Theorem. If f is a differentiable, one-to-one function defined on an interval I, and x_0 is a point of I at which $f'(x_0) \neq 0$, then f^{-1} is differentiable at $y_0 = f(x_0)$ and

$$\frac{df^{-1}}{dy}(y_0) = \frac{1}{\frac{df}{dx}(x_0)}, \text{ where } y_0 = f(x_0).$$

Use the theorem above to answer the following questions.

3. For which points in the interval $[-1,1]$ is $\sin^{-1}(y)$ differentiable? Justify your answer.

4. Show that $\dfrac{d}{dy}\sin^{-1}(y) = \dfrac{1}{\cos(\sin^{-1}(y))}$.

5. We can simplify the expression in the previous question using the Pythagorean identity $\sin^2(x)+\cos^2(x)=1$, which implies that $\cos(x)=\pm\sqrt{1-\sin^2(x)}$. Recalling that $x\in(-\frac{\pi}{2},\frac{\pi}{2})$, determine the correct sign to take.

6. Plug in your expression for $\cos(y)$ into the expression $\dfrac{d}{dy}\sin^{-1}(y)=\dfrac{1}{\cos(\sin^{-1}(y))}$ and simplify as much as possible. $\diamondsuit$ **Check your answer with your instructor before moving on.**

7. Use the relationship you found in question 2 to find $\frac{d}{dy}\cos^{-1}(y)$ and its domain without doing additional work. (Do not repeat all of the steps on the previous questions; your answer should be immediate.)

8. For which points in $(-\infty, \infty)$ is $\tan^{-1}(x)$ differentiable? Justify your answer.

9. Show that $\dfrac{d}{dy}\tan^{-1}(y) = \dfrac{1}{\sec^2(\tan^{-1}(y))}$.

10. What does the Pythagorean identity $\sin^2(x) + \cos^2(x) = 1$ become when you divide both sides by $\cos^2(x)$?

11. Calculate $\frac{d}{dy}\tan^{-1}(y)$ and use the identity you derived in 9 to simplify as much as possible.
$\diamondsuit$ **Check your answer with your instructor before moving on.**

12. Use the relationship you found in question 2 to find $\frac{d}{dy}\cot^{-1}(y)$ and its domain without doing additional work. (Do not repeat all of the steps on the previous questions; your answer should be immediate.)

13. For which points $|y| \geq 1$ is $\sec^{-1}(y)$ differentiable?

14. Show that $\dfrac{d}{dy}\sec^{-1}(y) = \dfrac{1}{y\tan(\sec^{-1}(y))}$.

15. By the identity $1+\tan^2(x) = \sec^2(x)$, $\tan(x) = \pm\sqrt{\sec^2(x)-1}$.

    (a) Determine the sign of $\tan(x) = \pm\sqrt{\sec^2(x)-1}$ when $x = \sec^{-1}(y)$ with $y > 1$. Then use this identity to simplify $\frac{d}{dy}\sec^{-1}(y)$.

(b) Determine the sign of $\tan(x) = \pm\sqrt{\sec^2(x) - 1}$ when $x = \sec^{-1}(y)$ with $y < -1$. Then use this identity to simplify $\frac{d}{dy}\sec^{-1}(y)$.

(c) Recall that

$$|y| = \begin{cases} y, & \text{if } y \geq 0, \\ -y, & \text{if } y \leq 0. \end{cases}$$

Use the absolute value function to combine your two expressions for $\frac{d}{dy}\sec^{-1}(y)$ into a single expression.

16. Use the relationship you found in question 2 to find $\frac{d}{dy}\csc^{-1}(y)$ and its domain without doing additional work. (Do not repeat all of the steps on the previous questions; your answer should be immediate.)

Summary

THEOREM 7.12 Derivatives of Inverse Trigonometric Functions

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1 - x^2}} \qquad \frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1 - x^2}}, \quad \text{for } -1 < x < 1$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1 + x^2} \qquad \frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1 + x^2}, \quad \text{for } -\infty < x < \infty$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2 - 1}} \qquad \frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2 - 1}}, \quad \text{for } |x| > 1$$